W-C-581b
November 8, 1963

SUPERSEDING
Int. Fed. Spec. W-C-00581a (GSA-FSS)
August 22, 1962 and
Fed. Spec. W-C-581
August 1, 1939

FEDERAL SPECIFICATION

CONDUIT AND FITTINGS; NONMETALLIC, RIGID, (BITUMINIZED HOMOGENEOUS FIBER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers conduit and fittings composed of a fibrous material thoroughly impregnated with a bituminous compound for use as a raceway for the installation of electrical wires and cables.

1.2 Classification.

1.2.1 *Types.* Conduit and fittings furnished under this specification shall be of the following types, as specified (see 6.1).

Type I—Designed to be encased in concrete when installed.

Type II—Designed for installation underground without encasement in concrete.

1.2.2 *Sizes.* Conduit and fittings furnished under this specification shall conform to standard commercial sizes, as specified (see 6.1).

1.2.3 *Kinds of fittings.* Fittings furnished under this specification shall conform to standard commercial kinds (couplings, elbows, bends, etc.), as specified (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitations for bid, form a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-3466—Packaging of Pipe, Wood, Machine-Banded, Inserted-Joint; and Pipe, Bituminized Fiber, for Overseas Shipment.

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Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Underwriters' Laboratories, Inc., Publication:

UL-543—Standard for Fiber Conduit.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

National Electrical Manufacturers Association Publication:

BC1-1959—Standard for Bituminized-Fiber Conduits and Fittings.

(Copies may be obtained from the National Electrical Manufacturers Association, 155 East 44th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 General. All material shall be new, suitable for the purpose intended, and free of any defects that affect the appearance or impair the serviceability of the product.

3.2 Detail. Conduit and fittings furnished under this specification shall conform to Standard BC1-1959 for Bituminized-Fiber Conduit and Fittings of the National Electrical Manufacturers Association (NEMA).

3.3 Fire and casualty hazards.

3.3.1 Each bidder shall submit to the contracting agency proof that the conduit and/or fittings he proposes to supply under this specification conform to the applicable requirements of Underwriters' Laboratories, Inc., Standard UL-543 for Fiber Conduit. The label or listing with re-examination of the Underwriters' Laboratories, Inc., may be accepted as evidence that the items conform to these requirements.

3.3.2 In lieu of the label or listing with re-examination the bidder may submit independent proof satisfactory to the contracting agency that the conduit and/or fittings conform to the applicable requirements, including methods of test, of UL-543.

3.3.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.4 Conduit length. The conduit length for each size shall be that which is standard with the manufacturer but shall not be less than 5 feet, plus 0, minus 1 inch. Short lengths of conduit shall not exceed 15 percent in any shipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, sampling for inspection and tests shall be in accordance with MIL-STD-105.

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4.3 Inspection. Unless otherwise specified, samples selected in accordance with 4.2 shall be inspected in accordance with NEMA Standard BC1-1959. The acceptable quality level (AQL) shall be 4 percent defective.

4.4 Tests. Samples selected in accordance with 4.2 shall be tested in accordance with NEMA Standard BC1-1959. The AQL shall be 4 percent defective. Any sample failing to pass one or more tests shall be considered defective.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the levels of preservation, packaging, and packing are defined in Fed. Std. No. 102).

5.1 Packing.

5.1.1 *Level A or B.* Conduit and fittings shall be packed in accordance with the requirements of MIL-P-3466.

5.1.2 *Level C.* Conduit and fittings shall be packed in accordance with standard commercial practice.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required in the contract or order, civil agency marking shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required in the contract or order, military agency marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Type of conduit and/or fitting (1.2.1).

(*c*) Size of conduit and/or fitting (1.2.2).

(*d*) Kind of fittings (1.2.3).

(*e*) Method of sampling, if other than specified (4.2).

(*f*) Inspection procedures, if other than specified (4.3).

(*g*) Level of packing required (5.1).

(*h*) Special marking, if required (5.2).

6.2 Fittings for fiber conduit (with the exception of couplings, one of which is supplied with each length of conduit) must be ordered in addition to conduit lengths.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Conduit or pipe, or conduit or pipe connections, fiber, bituminized, not otherwise indexed by name.
Carload minimum weight: 30,000 pounds.

Motor:

Conduit or pipe or connections or fittings, bituminized fiber.
Truckload minimum weight: 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:

Army—MO

Navy—YD

Air Force—ROM

Copies of this specification may be purchased for 60 cents each.

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